

AutoGen CAP Admission Counsellor AI

Setup, Deployment, and Local Execution Guide

This repository contains an automated intelligent system for parsing engineering cutoff data from Maharashtra state CAP round PDF documents and matching candidate percentiles to appropriate college and course options via a multi-agent AutoGen architecture.

1. System Requirements & Compatibility

- **Python Version:** Python 3.10 or 3.11 is recommended. (Minimum requirements dictate Python 3.8+).
- **Operating System:** Platform independent (Windows, macOS, or Linux).
- **Internet Connection:** Required to query the remote Groq LLM API backend during optimization steps.

2. Project Directory Layout

Before launching initialization or execution scripts, verify that your project structure matches the layout below. The source PDF data file must reside directly in your project root directory.

```
Counselor/
├── agentic_counsellor_app/
│   └── app.py # Main AutoGen application code
├── 2023ENGG_CAP1_CutOff.pdf # Source cut-off document (Required)
├── requirements.txt # Third-party dependencies
└── .gitignore # Git inclusion filter mappings
```

3. Installation Step-by-Step

Step 3.1: Create a Clean Virtual Environment

Open your command line interface or PowerShell at the project root directory and initialize an isolated workspace environment:

```
# On Windows (PowerShell)
python -m venv env
.\env\Scripts\Activate.ps1

# On macOS / Linux
python3 -m venv env
source env/bin/activate
```

Step 3.2: Install Package Dependencies

Install the streamlined, minimal requirement set using pip. All necessary underlying packages (like Pydantic and NumPy) will handle automatic resolution during download invocation:

```
pip install -r requirements.txt
```

4. Environment Variables Configuration

The application relies on a secure Groq API key token to communicate with the Llama and Mixtral model fallbacks. You must declare this token in your active terminal instance prior to launching the script execution command.

```
# On Windows (PowerShell)
$env:GROQ_API_KEY="your_actual_groq_api_key_here"

# On macOS / Linux
export GROQ_API_KEY="your_actual_groq_api_key_here"
```

Security Warning: Avoid hardcoding API strings or configuration credentials directly into your committed code tracked by Git. Always reference runtime tokens utilizing environment paths.

5. Execution Instructions

To run the intake portal interface, ensure your environment state is active and environment variables have been set, then invoke the runner application:

```
python . gentic_counsellor_app pp.py
```

Interacting with the Prompt Console

When triggered, you will see a console menu requesting input credentials. Provide any human-readable string containing percentile metrics, cast status information, and preferred courses:

```
Example Input:
"I scored 94.2 percentile, my category is OBC, and I am looking for CS or IT branches."
```

6. Automated Deliverables Output

Upon completion, the AutoGen workflow will produce a styled executive performance document matching the profile parameter parameters:

- **Output Filename:** `CAP_Round_Allotment_Preferences.xlsx`

- **Contents:** Up to 25 optimized, unique choice selections.
- **Formatting Themes:** Professional corporate navy blue headers, structured data widths, automatic gridlines, and built-in categorical reasoning logs.